DeSurv, a survival-supervised matrix factorization, identifies a replicable tumor–stroma prognostic architecture in pancreatic cancer

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Abstract

Pancreatic ductal adenocarcinoma (PDAC) outcome reflects malignant and stromal state, yet transcriptomic programs are typically discovered without outcome information and tested against survival afterward. We developed DeSurv, a survival-supervised matrix factorization that incorporates outcome into gene-program estimation and holds the gene weights fixed for scoring new cohorts. In pooled TCGA and CPTAC cohorts, DeSurv recovered a three-program tumor–stroma architecture whose dominant survival-aligned direction (D1) linked classical tumor identity with restCAF-like stroma. The frozen representation transferred across five external datasets spanning four independent cohorts, where the combined risk score remained prognostic (HR per SD, 1.42; 95% CI, 1.26–1.59), including after PurIST and DeCAF adjustment. Unsupervised factorization matched on program number and tuning budget did not recover D1, indicating that it arose from survival supervision, not model size or computational effort. In simulations with known truth, DeSurv more accurately recovered the genes defining the prognostic program when it contributed little to variance.

Pancreatic ductal adenocarcinoma (PDAC) is among the most lethal solid tumors[1]. Bulk transcriptomic studies have identified clinically relevant malignant and stromal states in PDAC: basal-like and classical tumor programs[2–4], and tumor-promoting proCAF and tumor-restraining restCAF stromal programs[5]. Single-cell and spatial profiling have further shown that such state heterogeneity extends across malignant, stromal, immune, and other microenvironmental compartments within individual tumors[6–8]. Whether a survival-aligned direction spanning tumor and stromal programs exists, and adds prognostic information beyond existing subtype classifications, remains unclear, in part because such programs are discovered without outcome information and tested against survival only afterward[2,9].

Answering that requires survival-annotated cohorts at scale, which in PDAC still means bulk transcriptomic profiling[10]. Bulk profiles, however, mix signals from malignant and microenvironmental cells[11]. Nonnegative matrix factorization (NMF) and related deconvolution methods separate these composite profiles into interpretable gene programs: weighted sets of co-varying genes that together form a representation of the tumor transcriptome[3,12–14]. In PDAC, such methods have yielded molecular subtypes[2,4,9], virtual microdissection of basal-like and classical malignant programs[3], and compartment-specific tumor–stroma deconvolution[5,14]. The practical challenge is therefore not simply to resolve interpretable gene programs. It is to let survival shape the programs as they are learned, to retain a decomposed representation rather than only a risk score, and to test those same programs, unchanged, across independent studies.

Standard NMF learns gene programs by minimizing reconstruction error, the mismatch between the measured expression matrix and the approximation rebuilt from the programs. Such variance-driven criteria favor patterns that capture prominent expression variation whether or not that variation is related to survival[15,16]. This is particularly important in PDAC, where bulk tumors contain substantial stromal, exocrine, and other non-malignant signals that can dominate transcriptomic variation and obscure less prominent but prognostically relevant programs[4,5,17,18]. Reconstruction alone also does not uniquely determine the recovered gene programs without additional constraints[19–21], which may contribute to cross-study variability among unsupervised PDAC subtype taxonomies[22]. Consequently, establishing which unsupervised programs are clinically relevant and transportable has required extensive multi-study refinement: in PDAC, the basal-like/classical dichotomy took roughly a decade of iterative subtyping, virtual microdissection, independent cohorts, single-cell analyses and prospective clinical profiling[2–4,8,10], and comparable work on stromal CAF subtypes is under way[5].

Conversely, survival-supervised models such as penalized Cox regression and supervised principal components[15,23] can identify outcome-associated expression patterns. Such models may predict survival without organizing the prognostic signal into biologically interpretable tumor and microenvironmental programs, which is the form in which PDAC subtypes, and the single-sample classifiers built from them, have been defined[4,5]. A risk score alone does not supply that substrate.

Here, we present DeSurv, a survival-supervised matrix factorization framework that integrates NMF with Cox regression so that survival helps shape gene programs as they are learned. By incorporating survival during factorization, DeSurv is designed to recover survival-aligned biological structure that may be obscured when other sources of expression variation dominate. A consensus across many restarts supplies the initialization for a final fit, after which the gene weights defining each program are held fixed, or frozen, allowing the same programs to be evaluated directly in independent cohorts without refitting the factorization. DeSurv therefore changes clinical relevance from a post hoc criterion used to select among unsupervised factors into a criterion that shapes the factors themselves, while making cross-study transportability a direct test of the same frozen gene programs. We evaluate DeSurv in PDAC across five external validation datasets spanning four independent patient cohorts, and use simulations in which the true underlying (latent) programs are known, to characterize when survival supervision improves recovery of prognostic programs.

In PDAC, DeSurv identifies a three-program tumor–stroma architecture that reflects joint cross-patient variation rather than spatial or cellular organization within individual tumors. We first compare it with NMF at higher rank (the number of programs the factorization extracts), with an unsupervised control matched on rank and tuning budget, and with conventional survival-supervised models, to isolate what survival supervision contributes to the representation itself (Supplementary Table S1). We then validate the frozen programs externally and ask whether the survival-aligned program is DeSurv-specific or reducible to established classifiers.

Results

DeSurv incorporates survival information into gene-program estimation

DeSurv learns gene programs by balancing reconstruction of the tumor transcriptome with association to patient survival. Unlike the conventional sequence of unsupervised factorization followed by post hoc clinical screening, survival directly shapes the gene programs during discovery. After training, the gene weights are fixed, allowing the same programs to be scored and tested in new tumors without refitting the factorization (Fig. 1A; Methods).

We trained DeSurv in pooled TCGA and CPTAC PDAC cohorts[24,25] ($n = 273$, 139 events). Settings were chosen by Bayesian optimization[26,27], a sequential search algorithm that proposes each new setting from the results of previous ones to optimize a pre-specified objective (Methods). Cross-validated concordance (the C-index, the probability that the model ranks a pair of patients in the correct survival order) was high across a broad region of intermediate supervision and moderate rank, declining toward the unsupervised $\alpha = 0$ boundary (Supplementary Fig. S1). Concordance changed little across the ranks examined, so we applied the one-standard-error rule, which takes the most parsimonious model within one standard error of the best, giving a three-program model ($k = 3$, $\alpha = 0.334$; Fig. 1B and Supplementary Fig. S4). By contrast, standard unsupervised NMF rank criteria (reconstruction residuals, cophenetic correlation and silhouette width[13,28]) gave conflicting choices in the same data (Supplementary Fig. S5). All model and hyperparameter selection was completed within the training cohorts before the programs were applied to external validation datasets.

Survival supervision changes the recovered representation at matched complexity

To determine how survival supervision altered the biological representation, we compared DeSurv with standard NMF (Lee multiplicative NMF, the field-standard implementation) at the same rank ($k = 3$), thereby holding model complexity constant. DeSurv resolved three distinct programs (D1–D3): D1, enriched for classical tumor and restCAF-like stromal signatures; D2, enriched for proCAF-like stroma; and D3, enriched for basal-like tumor programs (Fig. 2a). Together, these programs define a tumor–stroma prognostic architecture spanning classical/restCAF-like and basal-like/proCAF-like states across patients. The classical identity of D1 was independently supported in the COMPASS cohort, where D1 scores correlated with GATA6 RNA in situ hybridization (Spearman $\rho = 0.68$, $P < 0.001$, $n = 33$; Fig. 2e).

Standard NMF produced a different representation at the same rank (Fig. 2b). Its malignant factor (N1) was enriched for both classical and basal-like tumor programs, its microenvironmental factor (N3) for multiple stromal and immune programs[29,30], and a third factor (N2) for exocrine-associated variation. DeSurv instead separated basal-like tumor (D3) and proCAF-like stroma (D2) while recovering the distinct classical/restCAF-like survival-aligned D1 program. Both methods corresponded comparably to the activated-stroma and proCAF-like reference programs, the stromal distinctions bulk deconvolution conventionally resolves. They differed at the restraining end: among the displayed top-ranked genes, the restCAF and iCAF reference programs corresponded to a DeSurv program and to no standard NMF factor (Fig. 2a,b).

This difference reflected a selective reorganization rather than wholesale loss of expression structure. D1 accounted for the smallest share of the full model's reconstruction gain, 18.8% (Shapley shares[31]), yet carried essentially all of the survival signal (Fig. 2c). Omitting it lowered the Cox partial log-likelihood, the fit criterion of the Cox model, by 66.4, against 1.0 for D3 and 0.02 for D2, even though D3 accounted for 56% of it. Because D1 was estimated using these outcomes, its increment describes where the model placed the survival signal rather than testing whether that signal is real. The unsupervised factors were fit without survival, and the largest reconstructor among them, at 42% of the reconstruction gain, lowered the partial log-likelihood by only 1.16. The programs that best reconstruct the transcriptome need not be those most relevant to survival[18].

The principal NMF tumor and microenvironmental factors were largely retained as D3 ($\rho = 0.84$) and D2 ($\rho = 0.87$), whereas D1 correlated only weakly with every matched-rank NMF factor ($\rho \leq 0.36$; Spearman correlation of gene loadings over all analysed genes; Fig. 2d). Consistently, D1 was poorly reconstructable from the full set of NMF factors ($R^2 = 0.09$; Supplementary Table S3). DeSurv nonetheless reconstructed the analysis matrix only somewhat less completely than matched-rank standard NMF (R^2 0.75 versus 0.81, a difference of 6.0 percentage points; Supplementary Table S4). Survival supervision therefore preserved most of the variance-dominant tumor and stromal structure while resolving a distinct survival-aligned program, at a measurable cost in reconstruction.

Tuning effort does not appear to explain the difference. We repeated the comparison with the matched-budget unsupervised control, an unsupervised model at the same rank given DeSurv's optimizer, its tuning budget and its consensus initialization (starting values pooled over many restarts). It recovered D2 ($\rho = 0.86$) and D3 ($\rho = 0.75$) as before, but recovered D1 no better than the standard implementation did ($\rho \leq 0.24$).

Frozen survival-aligned programs retain prognostic relevance across independent untreated and treated PDAC cohorts

We next applied the frozen programs to five external validation datasets spanning four independent patient cohorts (Dijk, Moffitt, PACA-AU [array and RNA-seq], and Puleo; non-metastatic, treatment-naive; Supplementary Table S2). Each tumor was scored by projecting its expression onto the trained programs, restricted to their top-ranked genes ($Z = \widetilde{W}^\top X_{\text{new}}$; Methods). Combined analyses represented each patient once, giving 570 unique patients (388 deaths; Methods). The survival-aligned D1 program had protective point estimates in every validation dataset and was the only individual program associated with survival in the combined, cohort-stratified analysis (HR per SD 0.75 (0.68–0.83), P <0.001). D3 pointed toward higher risk in four of the five datasets but was not significant in any of them, nor combined (1.09 (0.98–1.21), $P = 0.10$). The proCAF-like D2 program was near-null (1.02 (0.92–1.12), $P = 0.76$) and more variable (Fig. 3a).

The frozen multivariable DeSurv linear predictor, the weighted sum of the three program scores (oriented so that higher values indicate higher risk), was associated with overall survival in a cohort-stratified analysis (pooled HR per SD = 1.42, 95% CI 1.26–1.59, $P = 5.4 \times 10^{-9}$; Table 1). Scaled Schoenfeld residuals showed the association attenuating over follow-up ($\chi^2 = 14.7$, 1 df, $P = 1.3 \times 10^{-4}$; Supplementary Information, Proportional-hazards diagnostic), so the pooled hazard ratio is a time-averaged summary rather than a constant multiplicative effect. The estimate did not depend on any single dataset (Fig. 3a).

Allowing cohort-specific slopes did not improve fit over a common slope (likelihood-ratio $\chi^2 = 1.5$, 3 df, $P = 0.69$; score standardized within cohort), and the four cohort estimates were descriptively homogeneous (Cochran $Q = 1.5$, $I^2 = 0\%$). Combined discrimination across cohorts was moderate (event-weighted C-index 0.61, 95% CI 0.58–0.65), with an integrated Brier score of 0.159 against 0.168 for a Kaplan–Meier reference (index of prediction accuracy 5.4%; Supplementary Information).

At the same complexity, unsupervised predictors were not associated with survival, whether tuned by DeSurv's own budget or not (Table 1).

We then asked whether comparable prognostic performance could be achieved without survival-supervised factorization. The two methods differ most in how their performance depends on rank (Supplementary Fig. S4). DeSurv reaches essentially its full cross-validated concordance at the smallest ranks examined and is flat thereafter, so added factors do not improve it. The unsupervised limit ($\alpha = 0$) starts well below DeSurv at low rank and climbs steadily as factors are added, closing most but not all of the gap by the largest rank examined. Across this range its cross-validated concordance does not reach the value DeSurv attains at $k = 3$, except at a single rank where the DeSurv curve itself dips. Unsupervised factorization therefore requires many more factors to approach the same cross-validated prognostic information.

The higher-rank solution also represented more established PDAC programs: coverage increased from 16 for DeSurv at $k = 3$ to 32 for the rank-optimized unsupervised control (the same pipeline at $\alpha = 0$ with rank selected by Bayesian optimization, $k = 7$), while reference-program multiplicity remained low (1.00 versus 1.125; Supplementary Figs. S6 and S7). Higher rank therefore primarily broadened biological representation rather than repeatedly mapping factors to the same programs. Survival supervision is not required to recover prognostic signal, but it concentrates that signal into a decomposition less than half the size (Table 1; Supplementary Table S6).

Table 1: Pooled external validation hazard ratios (per SD of the linear predictor) for DeSurv at $k = 3$, the matched-budget unsupervised control ($k = 3$, $\alpha = 0$) and the rank-optimized unsupervised control ($k = 7$, $\alpha = 0$), across five validation datasets spanning four independent patient cohorts (combined $n = 570$ unique patients, 388 deaths). Adjusted models include the PurIST tumor-intrinsic and DeCAF stromal molecular classifiers as covariates, with stratification by dataset, and are restricted to the $n = 570$ patients (388 deaths) with both classifier calls available. DeSurv and the rank-optimized control retain significant prognostic value after adjustment. Both unsupervised controls are the unsupervised limit of the same optimizer, run at $\alpha = 0$ with the identical consensus initialization and Bayesian-optimization budget given to the supervised model: the rank-optimized control at its own BO-selected rank and the matched-budget control held at DeSurv's rank, the latter serving as the tuning-budget control for the representational comparison in Fig. 2. The matched-rank comparator used for the factor-structure panels of Fig. 2 is standard NMF (Lee multiplicative NMF, the field-standard implementation) (Supplementary Table S6).

Method	Unadjusted HR per SD (95% CI), P	Adjusted HR per SD (95% CI), P
DeSurv ($k = 3$)	1.42 (1.26–1.59), < 0.001	1.28 (1.11–1.47), < 0.001
Matched-budget unsupervised control ($k = 3$, $\alpha = 0$)	1.04 (0.94–1.16), 0.428	0.98 (0.88–1.09), 0.705
Rank-optimized unsupervised control ($k = 7$, $\alpha = 0$)	1.44 (1.30–1.59), < 0.001	1.40 (1.22–1.62), < 0.001

The programs above were learned in treatment-naive, non-metastatic disease. Projected without retraining into independently treated PDAC cohorts, the survival-aligned D1 program remained associated with overall survival in treated metastatic disease under an arm-stratified analysis, and the proCAF-like D2 program increased after treatment (Fig. 4; Supplementary Information, Translational transportability).

The survival-aligned structure is not method-specific and is complementary to established PDAC classifiers

We next asked whether the dominant D1 direction simply recapitulated established PDAC subtype classifications. Across the pooled validation cohorts ($n = 570$, 388 events), binary PurIST and DeCAF classifications together explained 34% of the variance in the continuous D1 score, indicating substantial alignment with known tumor-intrinsic and stromal subtypes but incomplete overlap. This is an ordinary R^2 from a linear model with cohort fixed effects, fitted to D1 scores standardized within cohort. D1 remained associated with survival after simultaneous adjustment for both classifiers (HR per SD = 0.82, 95% CI 0.73–0.93, $P = 0.0014$). Adding D1 to a stratified Cox model with additive PurIST and DeCAF terms improved fit beyond those terms (partial likelihood-ratio $\chi^2 = 10.1$ on 1 df, $P = 0.0015$) and concordance from 0.592 to 0.622. In Puleo, the largest validation cohort, the DeSurv linear predictor remained prognostic after adjustment for standard clinical covariates (HR per SD = 1.33, 95% CI 1.14–1.55; Supplementary Table S5).

Because D1 spans tumor and stromal programs, we asked whether it tracks tumor purity. Where purity was measured on a full cohort the association was essentially unchanged (PACA-AU array, 1.44 unadjusted versus 1.45 adjusted; RNA-seq, 1.66 versus 1.65), arguing against a purity artifact (Supplementary Table S5).

The survival-aligned representation overlaps established tumor and stromal classifications without being reducible to them. The classifier analyses provide two complementary results. DeCAF classification alone explained 19% of the variance in D1, supporting alignment with DeCAF-defined stromal variation, whereas D1 remained prognostic after simultaneous adjustment for PurIST and DeCAF, showing that it is not reducible to those classifiers. Neither result establishes restCAF identity at the cellular level, which this bulk-expression analysis does not resolve (Supplementary Information, Stromal resolution: which reference programs are recovered from bulk).

The dominant D1 direction was also not specific to the DeSurv optimizer. Three independent survival-supervised approaches, in simple single-component form (supervised principal components, Cox partial least squares and sparse Cox), recovered patient scores strongly correlated with D1 in the training cohort ($|r| = 0.81$, 0.83 and 0.85, respectively). The leading unsupervised principal component was only weakly correlated with D1 ($|r| = 0.10$) and instead tracked higher-variance D2/D3 structure (Supplementary Information, The D1 axis is not specific to DeSurv). Thus, D1 reflects a survival-associated expression direction shared across supervised approaches rather than an idiosyncrasy of DeSurv.

Having established that D1 is not specific to DeSurv, we asked what DeSurv adds beyond recovering it. Independently tuned supervised principal components and penalized Cox achieved pooled external concordance comparable to DeSurv (C-index 0.60, 0.60 and 0.61, respectively) and captured the dominant D1-associated prognostic direction. Their selected genes, however, were consistently enriched for basal-like tumor programs but not proCAF, restCAF, immune or exocrine programs. Their risk scores were weakly correlated with the separately resolved proCAF-like D2 program (penalized Cox, $|r| = 0.06$; supervised PCA, $|r| = 0.14$; Fig. 3b). Although these scores retained modest stromal signal, they did not resolve it as a distinct stromal program (Supplementary Information). DeSurv instead places that direction within a fixed set of tumor and stromal programs that can be measured and interpreted separately, and applied unchanged to new cohorts.

DeSurv recovers prognostic gene programs when variance and prognosis diverge

We used simulations with known latent structure to characterize when survival supervision alters program recovery. In the primary scenario, the factor-specific genes driving survival contributed less transcriptomic variation than the shared outcome-neutral background signal ($p = 3{,}000$ genes, $n = 200$ samples, true $k = 3$; Methods, Simulation studies). DeSurv and unsupervised NMF ($\alpha = 0$, the unsupervised limit of the same pipeline) were tuned using the same cross-validation procedure, isolating the effect of survival supervision during factorization.

Under this setting, DeSurv more accurately recovered the genes defining the true prognostic program, whereas unsupervised NMF largely followed variance-dominant structure (Fig. 5a–b). The genes assigned to a recovered program, not only its ability to rank patients by risk, determine its subsequent biological interpretation and its potential use in candidate signatures or biomarker development. Across 100 replicates, median test-set C-index was 0.837 for DeSurv versus 0.724 for unsupervised NMF ($\Delta = 0.113$, paired Wilcoxon $P < 0.001$), and DeSurv recovered the true factorization rank ($k = 3$) more frequently (Fig. 5c).

The advantage depended on the relationship between transcriptomic variance and prognosis. It attenuated when prognostic and variance-dominant structure partially overlapped and disappeared under the null, where neither method identified prognostic signal (Supplementary Fig. S3). In the mixed setting, the advantage remained strongest for recovery of the factor-specific marker genes rather than shared background survival genes, supporting a specific effect of supervision on identifying the genes that define the prognostic program.

Discussion

DeSurv integrates survival into gene-program estimation while preserving a fixed basis that can be scored unchanged in new cohorts. In PDAC, this identified a dominant survival-aligned program, D1, linking classical tumor identity to restCAF-associated stromal biology, alongside separately resolved proCAF-like stromal and basal-like tumor programs. D1 was recovered de novo without subtype labels and remained prognostic after adjustment for the established PurIST and DeCAF classifications, so it is not reducible to them. Its classical tumor identity was supported by GATA6 RNA in situ hybridization. The trained programs then transferred without refitting across five external validation datasets spanning four independent patient cohorts. DeSurv's contribution was to place this replicable prognostic direction within a biologically decomposed tumor–stroma architecture. The advance is in how clinically relevant programs are discovered, not in how well they predict.

The distinction was not one of model size, tuning effort or predictive accuracy. An unsupervised model given DeSurv's rank, optimizer, tuning budget and consensus initialization still failed to recover D1. The rank-optimized unsupervised control and conventional supervised predictors each recovered the prognostic information while representing its biology differently. The former spanned a broader set of established programs across seven factors; the latter resolved a tumor-intrinsic axis without the stromal programs. In simulations, where the latent programs are known, DeSurv more accurately recovered the genes defining the true prognostic program.

This addresses a practical limitation of the conventional discovery sequence. When programs are learned without reference to outcome, their clinical relevance and correspondence across datasets must be established retrospectively through factor selection, biological annotation and cross-study matching. DeSurv does not eliminate the need for external validation, but makes that validation a direct test of programs already shaped by clinical relevance and held fixed across studies.

This follows from the structure of the DeSurv objective. Survival supervision acts directly on the gene-program matrix W, while the reconstruction objective requires those same programs to continue representing the observed transcriptome. Once learned, the gene weights are fixed, so the same programs can be scored in a new cohort without re-estimating the factorization. Empirically, supervision selectively reorganized rather than replaced the variance-driven representation: it retained the tumor and stromal structure carried by D3 and D2 while introducing a D1 program poorly reconstructable from the matched-rank NMF factors. Prior survival-representation approaches, including survival-supervised matrix factorization[32,33] and deep latent-variable models[34,35], have also integrated outcome information with learned representations, but generally do not preserve the same directly transferable gene-program decomposition (Supplementary Table S1).

Biologically, D1 represents the dominant survival-aligned program within the broader tumor–stroma architecture rather than the architecture itself. It links classical tumor identity with restCAF-like stromal biology. The additional D2 and D3 programs provide separately measurable stromal and tumor context around this direction, rather than necessarily representing independent components of the fitted risk score. This architecture is consistent with growing evidence that PDAC outcome reflects interactions between malignant lineage state and heterogeneous fibroblast and stromal programs[36–38].

Rank selection is an additional source of uncertainty in unsupervised decomposition. In the PDAC training data, commonly used NMF criteria gave conflicting rank choices, and increasing rank progressively extended correspondence to additional established signatures across more factors. Together with the non-uniqueness of reconstruction-based factorization, this rank dependence may contribute to variability among unsupervised molecular taxonomies across studies[2,3,9]. Survival supervision does not remove that non-uniqueness, and single DeSurv runs from different random starts recover appreciably different gene sets. What makes the reported decomposition a stable object is the consensus step, and independently seeded replicates of the full procedure recover it closely (Supplementary Information, Stability of the recovered gene programs). What survival supervision adds is an outcome-relevant criterion for choosing among representations when the objective is to identify clinically relevant programs.

Our simulations clarify when this approach is most useful. When the genes defining the prognostic program contributed relatively little to overall transcriptomic variation, survival supervision improved their recovery; this advantage diminished as prognostic and variance-dominant structure overlapped and disappeared when no survival signal was present. Correct program recovery matters beyond discrimination because the genes assigned to a program determine downstream pathway interpretation, candidate signatures and potential biomarker development.

DeSurv is also distinct from reference-based methods such as CIBERSORTx[39] and BayesPrism[40], which estimate cellular composition using external cell-state references. Instead, DeSurv learns outcome-aligned latent gene programs directly from bulk expression and could in principle be combined with reference-based cell-fraction estimates to refine their cellular attribution.

Several limitations define the scope of these findings. DeSurv currently uses a proportional-hazards survival objective and was evaluated here only in PDAC. Extensions to non-proportional or nonlinear survival relationships are natural methodological directions. More importantly, the DeSurv formulation is not PDAC-specific, but whether its representational advantage generalizes across cancer types remains unknown. Testing it will require multi-cancer benchmarking against unsupervised decompositions and against strong penalized-regression baselines, which remain difficult to outperform consistently for out-of-sample prognosis[41].

The treated-cohort analyses reported above are exploratory. Within the O'Kane/COMPASS cohort the individual treatment arms were too small to show a consistent treatment-specific survival pattern, whereas the prognostic association held in the better-powered arm-stratified analysis. Whether these programs predict differential treatment benefit therefore remains unresolved and will require adequately powered, prespecified treatment-by-program interaction analyses.

The tumor–stroma architecture described here is a bulk-tissue phenomenon. It reflects cross-patient covariation of tumor- and stroma-associated transcriptional programs, not cellular co-expression or spatial adjacency within individual tumors. The biological identity of the survival-aligned D1 program is nevertheless supported by the purity-adjustment, DeCAF-adjustment, cross-method recovery and GATA6 evidence above. Mapping these bulk programs onto specific cellular states is not a matter of projecting the same gene weights into single-cell or spatial data. Most of the genes defining each program go undetected in any individual cell, so per-cell projection scores partly reflect detection depth rather than program activity, and multicellular spatial spots mix the compartments the programs are meant to separate. Establishing the cellular correlates of these programs will therefore require clinically annotated single-cell and spatial cohorts together with scoring methods designed for sparse measurement.

More broadly, unsupervised transcriptomic discovery is optimized to recover what varies most, and clinically supervised prediction to recover what predicts outcome. These need not be the same biological structure. DeSurv addresses this gap by incorporating survival into program discovery while retaining an explicitly decomposed gene-program representation. In PDAC, this revealed a replicable tumor–stroma prognostic architecture that could be transferred across cohorts, supported by independent biological evidence and interrogated beyond a risk score alone. By making clinical relevance part of discovery and cross-study transportability a test of the same frozen gene programs, it complements unsupervised deconvolution and conventional supervised prediction without depending on superior predictive accuracy.

Methods

Problem formulation and notation

Let $X \in \mathbb{R}_{\geq 0}^{p \times n}$ denote the nonnegative gene expression matrix. DeSurv approximates $X \approx WH$, where $W \in \mathbb{R}_{\geq 0}^{p \times k}$ contains nonnegative gene programs and $H \in \mathbb{R}_{\geq 0}^{k \times n}$ contains sample loadings. Additionally, let $y \in \mathbb{R}_{>0}^{n}$ denote patient survival times, $\delta \in \{0,1\}^n$ the event indicators ($\delta_i = 1$ for an observed event, $\delta_i = 0$ for censoring), and $\beta \in \mathbb{R}^k$ the Cox regression coefficients. Survival outcomes are modeled through a Cox proportional hazards model with factor scores $Z = W^\top X \in \mathbb{R}^{k \times n}$: for sample i with projected score $z_i = W^\top x_i$, the hazard is $h(t \mid x_i) = h_0(t) \exp(z_i^\top \beta)$, where h_0 is the baseline hazard. Defining factor scores by projection onto the learned gene programs ($Z = W^\top X$) rather than by the inferred loadings H lets the same gene weights score new cohorts without refitting, the property we exploit for external validation.

The DeSurv Model

DeSurv integrates NMF with penalized Cox regression to identify gene programs associated with survival outcomes. The method operates in a survival-supervised framework: the reconstruction loss penalizes deviation from the observed expression data, while the survival term directs gene programs toward outcome-relevant structure. The parameter $\alpha \in [0, 1)$ controls the relative contribution of each objective.

The joint objective is

$$\mathcal{L}(W, H, \beta) = (1 - \alpha)\, \mathcal{L}_{\mathrm{NMF}}(W, H) - \alpha\, \mathcal{L}_{\mathrm{Cox}}(W, \beta), \tag{1}$$

where $\mathcal{L}_{\mathrm{NMF}}(W, H)$ is the NMF reconstruction error (with an optional ℓ_2 penalty on H, set to zero in all analyses reported here) and $\mathcal{L}_{\mathrm{Cox}}(W, \beta)$ is the elastic-net penalized partial log-likelihood. Both terms are normalized before weighting, the reconstruction error by $\|X\|_F^2$ and the partial log-likelihood by $n_{\mathrm{event}}/2$, and the W update balances the magnitudes of the two gradients before α sets their relative contribution (Supplementary Information, Section 2). A key design choice is where survival supervision enters the factorization. Because factor scores are defined as $Z = W^\top X$, the Cox partial likelihood $\mathcal{L}_{\mathrm{Cox}}$ is a direct function of W and β, and the survival gradient acts explicitly on the gene program matrix W. The sample-level loadings H enter only through the reconstruction term and receive no survival gradient, preserving their interpretation as mixture coefficients[12,42]. This construction makes clinical outcome part of program estimation rather than a post hoc filter applied to an unsupervised decomposition.

When $\alpha = 0$, DeSurv reduces to its unsupervised limit, and β is fit post hoc by penalized Coxnet on the projected scores $Z = W^\top X$, used only for scoring and cross-validation without affecting that fit; across the pipeline, the survival-tuned n_{top} still enters the consensus initialization of the final $\alpha = 0$ model (Supplementary Information, Bayesian Optimization). The Cox component also supports optional adjustment for sample-level covariates (for example, stage or grade); the analyses reported here use factor scores alone.

Optimization alternates updates for H, W, and β (Supplementary Information, Section 2), with backtracking used for proposed W updates. The reconstruction term is minimized using multiplicative updates that preserve nonnegativity; the supervision term enters W's gradient as a Cox partial-likelihood derivative; and β is updated by Coxnet coordinate descent on the current factor scores $Z = W^\top X$ with elastic-net penalty parameters (λ, ξ).

Because the implemented algorithm combines normalization, gradient balancing and a Coxnet approximation, we assessed optimization behavior empirically rather than relying on a formal convergence guarantee. Across independent random initializations of the reported PDAC model, all 100 runs terminated on the stopping criterion and the fitted objective decreased monotonically to numerically stable solutions (Supplementary Information, Empirical optimization stability). Because objective convergence does not imply that the same programs are recovered each time, we separately assessed program-level stability. Individual restarts are dispersed, whereas the reported consensus procedure recovers the same decomposition closely across independently seeded replicates (Supplementary Information, Stability of the recovered gene programs). Closed-form update equations are given in Supplementary Information, Sections 1–2, and Bayesian-optimization control parameters in the Bayesian Optimization section.

Hyperparameter selection and cross-validation

Hyperparameters $(k, \alpha, \lambda, \xi, n_{\text{top}})$ were selected by maximizing the cross-validated Harrell C-index using Bayesian optimization (BO)[26,27]. The final rank k was chosen by a one-standard-error rule: for each rank, the Bayesian-optimization configuration with the highest observed mean cross-validated C-index was retained, the threshold was set one fold-level standard error below the overall best configuration's mean, and the smallest rank whose retained configuration met the threshold was selected; the remaining hyperparameters were those of the overall best configuration (Supplementary Information, Bayesian Optimization). All model selection was performed entirely within the training data using cross-validation; no validation cohort data were used during any stage of tuning, and out-of-sample generalization was assessed only on the external validation cohorts. Each fold was trained using multiple random initializations, and fold-level performance was defined as the average C-index across initializations; the consensus initialization described next was used only for the final model and is not reproduced within each fold. For stability, we used a consensus-based initialization for the final model, aggregating multiple DeSurv runs into a gene-gene co-occurrence matrix and constructing an initialization W_0 from the resulting clusters (Supplementary Information, Consensus initialization for the final model). Before projection, the gene support was restricted to the union across factors of each program's BO-selected top genes, ranked by factor specificity (Supplementary Information, Section 8), giving the truncated matrix $\widetilde{W}$ that was held fixed for all validation cohorts. Two scoring conventions follow from this support and are used for different quantities. Factor-level scores retain all k columns, $Z = \widetilde{W}^{\top} X_{\text{new}}$, and are used for the per-factor and pooled survival analyses. Where a single risk score is required, as in cross-validated tuning, the columns are first collapsed to $\theta = \widetilde{W}\beta$ and θ is scaled to unit ℓ_2 norm before projection. The two differ only by a positive scalar and so give identical concordance. The normalization applies to the collapsed score rather than to individual columns of W. In PDAC, BO selected $n_{\text{top}} = 270$ top genes per factor. The ridge penalty on the loadings (H) was held fixed at 0 in all analyses and was not tuned; factor scaling was instead controlled directly by column-normalization of W during optimization.

Because the goal is prognostic factorization, DeSurv selects k using the criterion that directly measures prognostic performance (cross-validated C-index) rather than the unsupervised diagnostics commonly used for NMF rank selection[13,28]. In the PDAC training data, standard NMF diagnostics (reconstruction residuals, cophenetic correlation, mean silhouette width) yielded inconsistent guidance across candidate ranks, a known limitation of unsupervised rank-selection criteria that can disagree on the same dataset (Supplementary Fig. S5). Cross-validated concordance varied little across candidate ranks, with no clear optimum anywhere between $k = 2$ and $k = 12$ and a total spread across that range of about two standard errors (Supplementary Fig. S4). The one-standard-error rule applied to joint BO over rank and supervision strength accordingly selected the parsimonious $k = 3$ at $\alpha = 0.334$. Biological support for the three-program solution comes from reference-program correspondence and the GATA6 comparison rather than from a sharp statistical optimum.

External validation cohorts were scored by projecting new datasets onto the learned programs via $Z = \widetilde{W}^{\top} X_{\text{new}}$, and survival associations were evaluated using Harrell's C-index, hazard ratios and log-rank statistics. Harrell's C-index is rank-based and assesses discrimination only, not calibration or time-dependent prediction error, so we additionally report a proper score as a sensitivity check. This is the integrated Brier score and its index of prediction accuracy relative to a Kaplan–Meier reference (Supplementary Information, Proper-scoring sensitivity of the external validation).

Because the survival-supervised information lives in W rather than H, scoring is a pure matrix projection requiring no retraining or validation survival information. Related supervised-NMF methods instead place the survival term on the sample side: CoxNMF maximizes a Cox partial likelihood whose covariates are the columns of H[32], and SurvNMF jointly learns the factorization and a Cox model on the sample-side latent matrix, which its own transposed notation writes as W[33]. Because the risk score under either formulation is a function of the sample loadings rather than of the gene-program basis, scoring an external cohort requires re-estimating those loadings; SurvNMF obtains them for held-out samples by a constrained least-squares fit (Supplementary Table S1). Validation survival outcomes were therefore used only for downstream performance evaluation, so reported external hazard ratios and log-rank statistics reflect purely out-of-sample generalization. Further training and validation details appear in Supplementary Information, Part I.

To characterize the biological identity of the learned factors, we quantified how strongly each factor's gene loadings aligned with independently defined PDAC tumor and microenvironmental gene programs. For each factor, we correlated its gene loadings with binary membership indicators for each reference program (Spearman correlation computed over the union of the top 50 genes of all factors of that method, intersected with the reference-program gene universe; this differs from the factor-to-factor loading correlations, which use all analysed genes), annotating Benjamini–Hochberg-adjusted P values within each factor as descriptive marks, because the gene universe is itself selected by loading. The same analysis was applied to DeSurv and to standard NMF fit at the same rank. For display, reference programs whose Spearman correlation exceeded 0.2 with at least one factor of either method are shown; the same threshold defines correspondence in the coverage analysis (Supplementary Information, Reference-program coverage, edges and multiplicity), as the same row set in both panels of the main figure (full details in Supplementary Information).

The matched-rank comparator shown in the figure is Lee multiplicative NMF (`NMF::nmf`), the field-standard implementation, which does not receive DeSurv's consensus initialization or hyperparameter tuning. We therefore also fit a matched-budget unsupervised control, which removes supervision and holds every other component of the DeSurv procedure fixed. Bayesian optimization was run with α pinned to 0 and the rank pinned to DeSurv's selected k, using the same optimizer, search budget, gene filter and cross-validation folds. The selected model was then fit by the same 100-restart consensus procedure. This control isolates the contribution of survival supervision while holding rank, optimization procedure, tuning budget, gene filtering and cross-validation folds fixed. A further sensitivity analysis instead held DeSurv's own λ, ξ and n_{top} fixed and varied only α. Both controls are reported in Supplementary Information, Matched-budget unsupervised control at the same rank.

To corroborate the classical component of D1 with an independent assay, we evaluated the association between projected D1 scores and GATA6 RNA in situ hybridization measurements in the COMPASS cohort among the 33 tumors with both measurements, using Spearman correlation. Because *GATA6* is itself in the 810-gene projection support (805 of these genes were measured in the COMPASS expression data and contribute to the projected scores), part of this correlation could reflect the score being built from the assayed gene. We therefore repeated the comparison with *GATA6* removed from the projection, and the correlation was essentially unchanged (Spearman $\rho = 0.67$ versus 0.68), indicating the association is not an artifact of self-correlation.

Supervised comparator analyses

To distinguish prognostic discrimination from biological representation, we independently fit supervised principal components (superpc; cross-validated screening threshold and component number) and lasso-penalized Cox regression (glmnet; λ_{min}) using the same 1,970-gene training universe as DeSurv. Each method performed feature selection and tuning using the training data only, after which the fitted model was frozen and applied to rank-transformed validation cohorts without refitting.

The two comparators select features by different mechanisms, and neither matches DeSurv's. The penalized Cox comparator uses a pure lasso penalty (`glmnet` with $\alpha = 1$), whereas supervised principal components applies no penalty at all, instead screening genes by a cross-validated feature-score threshold and extracting a cross-validated number of components. DeSurv's internal Cox step uses an elastic net whose Bayesian-optimization-selected mixing parameter ($\xi = 0.056$) sits close to ridge. Both comparators retain far fewer genes than DeSurv's 810-gene support (across resampling seeds, sparse Cox selected a median of 14 genes, range 9–33, and supervised principal components 39, range 5–222), and both were tuned by their standard method-specific cross-validation rather than a budget matched to DeSurv; the comparison therefore characterizes the representations each approach recovers rather than benchmarking tuned predictive performance.

No matched-budget control was run for these supervised comparators, as it was for the unsupervised arm, so the comparison with them concerns what each representation resolves, not tuned predictive performance. Correspondence between each supervised risk score and the DeSurv programs D1–D3 was quantified in the training cohort using absolute Spearman correlation. Additional analyses of selected-gene enrichment and compartment attribution are described in the Supplementary Information.

Simulation studies

Simulation studies were conducted to assess recovery of prognostic latent structure and survival prediction under controlled conditions. Gene expression data were generated from a nonnegative factor model $X = WH$, where gene loadings W comprised three gene classes: marker genes, background genes, and noise genes. Marker genes were simulated to load strongly on a single factor and weakly on others, background genes to load strongly across all factors, and noise genes to have uniformly low loadings. Specifically, marker gene primary loadings were drawn from $\text{Gamma}(\text{shape} = 3, \text{rate} = 0.8)$ and cross-loadings onto other factors from $\text{Gamma}(1, 20)$; background gene loadings from $\text{Gamma}(2, 1)$ across all factors; and noise gene loadings from $\text{Gamma}(1, 20)$. Sample-level factor activities H were drawn from $\text{Gamma}(2, 2)$. Each dataset comprised $G = 3{,}000$ genes, $N = 200$ samples, and $K = 3$ factors, with 150 marker genes per factor and 500 shared background genes. The three factors are exchangeable in this design: the prognostic factor is not generated with lower variance than the other two. The variance-dominant, outcome-neutral signal is the shared background block, which loads on every factor, so the factor-specific marker genes that carry survival contribute less of the total expression variation than that block. Cox regression coefficients were $\beta = (2, 0, 0)^\top$ in the prognostic scenario, $\beta = 0$ in the null scenario, and $\beta = (2, 0, 0)^\top$ in the mixed scenario where survival risk was driven by 300 genes comprising the 150 factor-1 markers and 150 randomly sampled background genes, creating partial overlap between the prognostic signal and variance-dominant outcome-neutral programs.

Survival times were generated from an exponential distribution. True factor scores $Z_{\text{true}} = X^\top W_{\text{true}}$, where W_{true} retained marker gene loadings for their corresponding factors and was zero otherwise, were standardized column-wise; the linear predictor was $\eta = Z_{\text{true}}\beta$; event times were drawn as $T \sim \text{Exponential}(\lambda_0 e^\eta)$ with $\lambda_0 = 0.05$, and censoring times independently as $\text{Exponential}(0.02)$. Each scenario comprised 100 replicates, each with a held-out test split. Each dataset was analyzed using DeSurv and unsupervised NMF ($\alpha = 0$, the unsupervised limit of the same pipeline) followed by Cox regression on inferred factors, both tuned using cross-validated concordance index via

Bayesian optimization. Performance was summarized across repeated simulation replicates using test-set concordance and matched-factor precision of prognostic-gene recovery.

Matched-factor precision was quantified separately for each survival-driving factor. We first matched each true program to the learned factor whose top-ranked genes had the highest precision against it. Precision was then the proportion of that learned factor's top-n_{top} genes belonging to the ground-truth prognostic gene set, with genes ranked by factor-specificity score and n_{top} set to the Bayesian-optimization-selected signature size. We report the average across survival-driving factors. Because n_{top} is selected separately for each method, we also report global marker recall, the fraction of the true survival-gene set recovered anywhere in the union of all learned factors' top-n_{top} gene sets, which does not depend on which learned factor carries the genes (Supplementary Information). In the primary scenario the target comprised the 150 factor-specific marker genes driving survival ($\beta = (2, 0, 0)$); in the mixed scenario, marker-only and full survival-gene definitions of the target were evaluated separately.

We considered three simulation scenarios: a prognostic scenario in which prognostic programs explain low variance relative to outcome-neutral signals, a null scenario with no survival signal, and a mixed scenario where prognostic and variance-dominant programs partially overlap.

Real-world datasets

We analyzed publicly available and controlled-access RNA sequencing (RNA-seq) and microarray cohorts of PDAC with corresponding overall survival outcomes. Eligible samples were primary PDAC tumors with a measured bulk expression profile and valid overall-survival annotation, subject to cohort-specific quality-control and exclusion criteria detailed in Supplementary Information, Section 6; accrual and follow-up periods are as defined in each cohort's original publication. Gene expression matrices were analyzed on a log scale: RNA-seq cohorts as $\log_2(\text{TPM} + 1)$; microarray cohorts in platform-normalized log-scale intensities. For each training cohort separately, genes were ranked by mean expression and by variance independently, and the top 3,000 genes with the smallest sum of the mean-expression and variance ranks were retained, selecting genes that are both highly expressed and highly variable. The intersection across training cohorts yielded 1,970 genes used for all analyses. Validation datasets were restricted to this same gene set. Survival times and censoring indicators were taken from the associated clinical annotations. No missing-data imputation was performed: all analyses used complete cases, and analyses adjusted for the PurIST and DeCAF classifiers were restricted to samples with available classifier calls. Of the seven PDAC expression datasets we considered, two were used for training (TCGA and CPTAC) and five for external validation (Dijk, Moffitt, PACA-AU array, PACA-AU RNA-seq, and Puleo); because the two PACA-AU platforms profile one patient cohort, these five validation datasets span four independent patient cohorts. To harmonize differences in scale across cohorts, filtered gene expression data were within-sample rank transformed before model training (ranks are inherently nonnegative, preserving NMF compatibility). Within-sample ranking reduces platform- and library-size differences in expression scale that would otherwise dominate cross-cohort projection. More details about the datasets can be found in Supplementary Information, Section 6.

Per-cohort participant flow is summarized in Supplementary Table S2. Of 321 pooled training-cohort samples (TCGA-PAAD, $n = 181$; CPTAC-3, $n = 140$), 48 were excluded for missing survival time, missing event indicator, missing tumor grade (a data-loading filter applied to the analytic training cohort; grade is not used by the model), or quality-control failure, yielding 273 analytic training samples (TCGA-PAAD 144, CPTAC-3 129; 139 events). Of 979 pooled validation-cohort samples, 363 were excluded for non-PDAC histology, non-tumor tissue, or missing survival annotation, yielding 616 analytic validation profiles across five platform datasets: Dijk ($n = 90$, 81 events), Moffitt GEO array ($n = 123$, 83), PACA-AU array ($n = 63$, 38), PACA-AU RNA-seq ($n = 52$, 31), and Puleo array ($n = 288$, 181).

PACA-AU included partially overlapping array and RNA-seq datasets (46 shared patients). Platform-specific analyses retained each dataset separately; for analyses combining validation datasets, duplicate patient accessions were represented once, preferentially retaining the RNA-seq profile and excluding the corresponding array profile, consistent with the preprocessing rule used in the previously published DeCAF analysis[5]. This yielded a combined validation population of 570 unique patients (388 deaths) spanning four independent patient cohorts.

Treated-cohort analyses

Two independently treated PDAC cohorts were used only for projection of the frozen programs (Supplementary Information, Translational transportability). In the O'Kane/COMPASS cohort of treated metastatic PDAC ($n = 173$ patients, 123 deaths; laser-capture microdissected, epithelial-enriched expression), the projected D1 score was tested in a Cox model stratified by treatment arm, with arm-specific estimates and a treatment-by-D1 interaction test reported descriptively; no estimate is pooled across arms. In the Linehan cohort of borderline-resectable or locally advanced disease, 29 patients with paired pre- and post-treatment biopsies were compared on the projected D2 score by a paired Wilcoxon signed-rank test; this analysis concerns program level, not survival. Both analyses were exploratory and not preregistered.

Use of large language models

During preparation of this manuscript, the authors used a large language model (Claude; Anthropic) to assist with language editing and with code for generating and formatting the display items. All such output was directed, reviewed, and verified by the authors, who take full responsibility for the work. The model was not used to design the study, develop the DeSurv methodology, or interpret the findings, and it does not meet the criteria for authorship.

Data Availability

The training and validation datasets used in this study are publicly available or accessible through controlled-access repositories, and are de-identified; no Institutional Review Board (IRB) approval was required for their analysis. Training data were obtained from the Genomic Data Commons: TCGA-PAAD[43] and CPTAC-3[44]. Validation cohorts were obtained from ArrayExpress (Dijk, E-MTAB-6830[45]; Puleo, E-MTAB-6134[46]), the Gene Expression Omnibus (GEO; Moffitt, GSE71729[47]), and the International Cancer Genome Consortium (ICGC): the PACA-AU array cohort is publicly available from the Gene Expression Omnibus (GSE36924), and the PACA-AU RNA-seq cohort from the European Genome–phenome Archive (EGA) study EGAS00001000154 under controlled access[48]. Expression data and molecular classifications (PurIST, DeCAF) for the validation cohorts were originally compiled for the DeCAF study[5]; cohort details and sample sizes after quality control are summarized in Supplementary Information, Section 6. The Linehan FOLFIRINOX$_{\pm}$CCR2-inhibitor series, used for the paired pre- and post-treatment comparison (Fig. 4b), and the O'Kane/COMPASS treated metastatic PDAC cohort used both for the D1 prognostic-transportability survival analysis (Fig. 4a) and, in the subset of 33 tumors with matched staining, for the GATA6 RNA in situ hybridization comparison (main-text Fig. 2e), are restricted-access clinical-trial datasets that are not publicly deposited; raw data are available from the respective study investigators under a data-use agreement, subject to the governing trial consents and institutional approvals. To preserve reproducibility without redistributing restricted data, the derived per-cohort summary statistics reported in the main text and Supplementary Information are provided in the reproducibility repository (file `results/treated_cohort_stats.rds`), so that every treated-cohort value can be inspected and verified. Code used to generate the COMPASS D1–GATA6 comparison (main-text Fig. 2e) is also provided in the reproducibility repository; reproduction of the patient-level analysis requires authorized access to the COMPASS data.

Code Availability

All analyses were performed in R (version 4.6.0; R Core Team) using DeSurv (v1.0.1; this work), `NMF`[42], `glmnet`[23], `survival`, `ggplot2`, and `cowplot`; Bayesian optimization for hyperparameter selection was implemented within DeSurv. The DeSurv R package is available at https://github.com/rashidlab/DeSurv (archived at Zenodo, DOI `10.5281/zenodo.20150929`). Reproducibility code, processed data, and the exact session-info file are available at https://github.com/rashidlab/DeSurv-paper (archived at Zenodo, DOI `10.5281/zenodo.20150946`); this includes `code/11_treated_cohort_analysis.R`, which generates the treated-cohort summary statistics above.

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Inclusion and Diversity. The patient cohorts analyzed in this study were curated by the original investigators or consortia (TCGA, CPTAC, ICGC, and the original Moffitt, Dijk, and Puleo studies) and our access was limited to the de-identified expression and survival data they provide. Sex was recorded in every cohort and was included as a covariate in the clinical-adjustment analyses (Supplementary Table S5), where the prognostic association of the DeSurv linear predictor was unchanged. We did not conduct sex-stratified analyses, because the study was not powered for subgroup comparisons within individual cohorts, and we did not stratify by race or ethnicity because those annotations were not consistently available across cohorts. Whether the prognostic relevance of these gene programs differs by sex, race, or ethnicity remains an important question for adequately powered, more comprehensively annotated cohorts.

Author Contributions

Conceptualization: A.M.Y., N.U.R. Methodology: A.M.Y., A.Y., D.L., N.U.R. Software: A.M.Y. Formal analysis: A.M.Y. Investigation: A.M.Y. Data curation: A.M.Y., X.L.P. Visualization: A.M.Y. Writing – original draft: A.M.Y., N.U.R. Writing – review and editing: A.M.Y., A.Y., X.L.P., J.J.Y., D.L., N.U.R. Resources (PDAC domain expertise and biological interpretation): X.L.P., J.J.Y. Supervision: N.U.R. Funding acquisition: A.M.Y., N.U.R. All authors reviewed and approved the final manuscript.

Competing Interests

N.U.R. and J.J.Y. are named as inventors on US patents US20220127676A1 and US20250066856A1, which relate to pancreatic cancer subtype classification. The PurIST and DeCAF classifiers were used only as published, independently derived comparators: their subtype calls were obtained with the released classifiers and included as fixed covariate annotations, and the adjustment analyses using these calls are objective and reproducible from the deposited code. They are not components of DeSurv. The other authors declare no competing interests.

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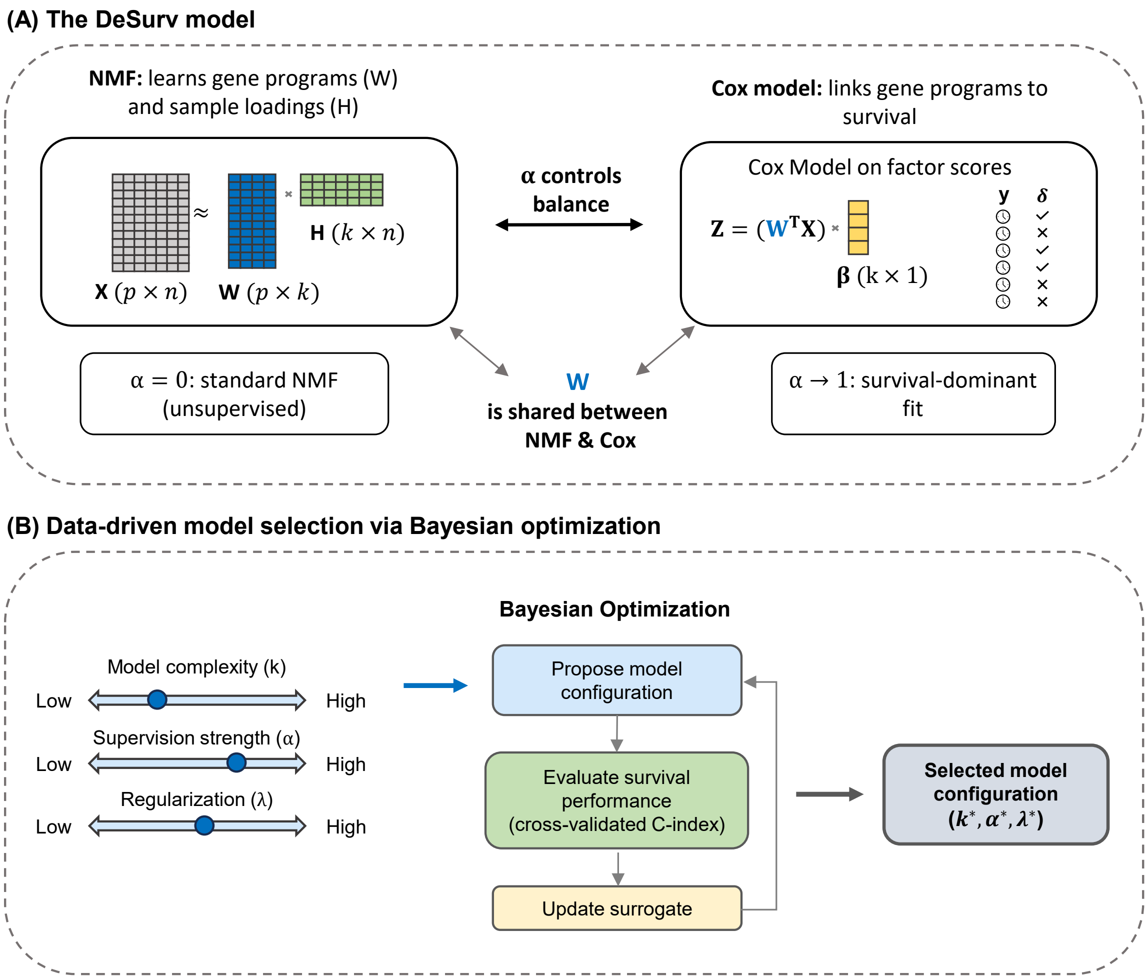

Figure 1: Overview of the DeSurv framework. (A) DeSurv jointly optimizes NMF reconstruction ($X \approx WH$) and Cox proportional hazards likelihood. The gene program matrix W is shared between objectives: factor scores $Z = W^\top X$ serve as covariates in the Cox model with coefficients β. A supervision parameter $\alpha \in [0, 1)$ controls the balance between reconstruction fidelity ($\alpha = 0$, the unsupervised limit) and survival prediction (α close to 1). (B) The factorization rank (k), supervision strength (α), and regularization (λ) are selected via Bayesian optimization over cross-validated concordance index. Survival supervision acts on the gene-program matrix W during discovery, rather than being applied only after an unsupervised factorization has been completed; following training-only model selection, the resulting gene weights are fixed for direct evaluation in external cohorts.

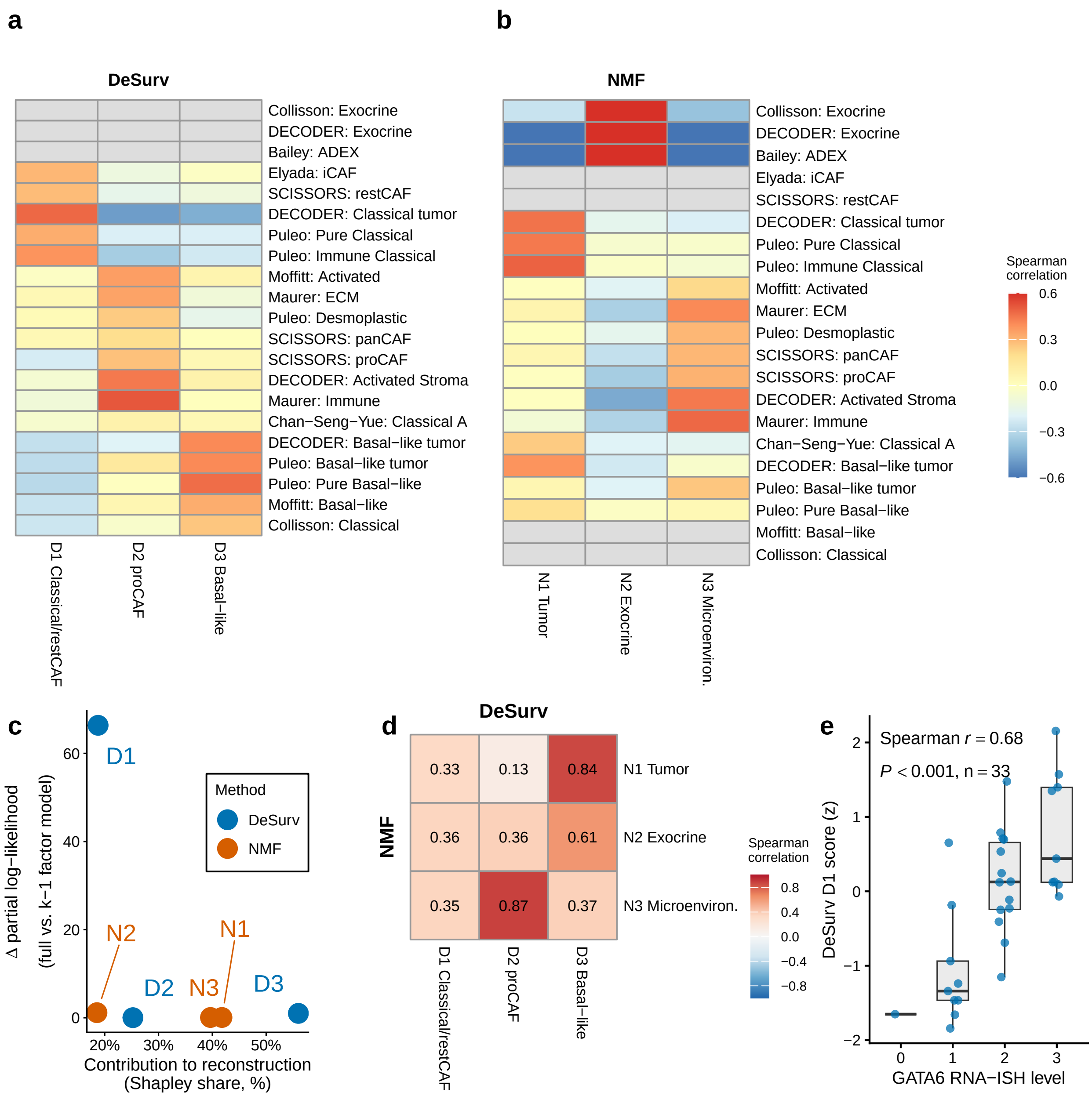

Figure 2: Survival supervision produces different factor structures in PDAC (TCGA + CPTAC training cohort, $n = 273$ patients, 139 events). (a,b) Spearman correlations between factor gene rankings (top 50 genes per factor) and established PDAC gene programs for DeSurv (a) and standard NMF (b) at $k = 3$. Both panels display the same rows, the union of reference programs correlating at $r > 0.2$ with at least one factor of either method; grey cells denote programs with none of their genes in that method's displayed gene space, where the correlation is not computable. The rows labelled SCISSORS: restCAF and SCISSORS: proCAF display the SCISSORS iCAF and myCAF gene sets under the naming used by the DeCAF classification (Supplementary Information). (c) Per-factor reconstruction contribution (normalized Shapley share, summing to 100%) versus survival contribution (Δ Cox partial log-likelihood on factor removal). Because D1 was estimated using these survival outcomes, its increment describes where the model placed the survival signal rather than testing whether that signal is real. (d) Pairwise correspondence between NMF and DeSurv factors (Spearman correlation of W-matrix gene loadings over all genes). (e) DeSurv D1 score versus GATA6 RNA in situ hybridization level in the COMPASS cohort (Spearman correlation; $n = 33$).

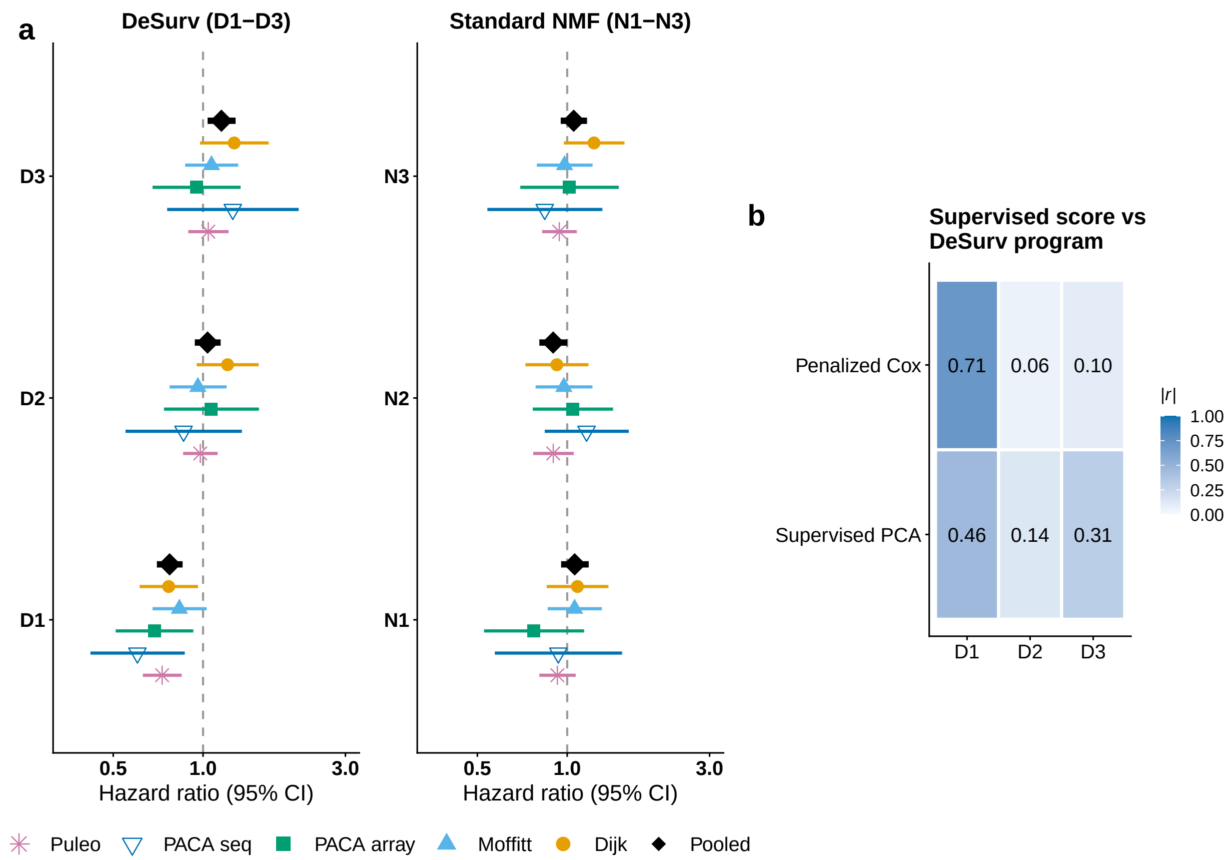

Figure 3: Survival-aligned programs generalize across independent PDAC patient cohorts (five external validation datasets: Dijk, Moffitt GEO array, PACA-AU array, PACA-AU RNA-seq and Puleo; combined $n = 570$ unique patients, 388 deaths, with PACA-AU patients profiled on both platforms represented once via RNA-seq). (a) Forest plot of per-dataset univariate hazard ratios (95% CIs) for each factor; the platform-specific PACA-AU array and RNA-seq estimates are shown separately, and the black diamonds denote the combined estimate over unique patients (per-factor cohort-stratified Cox on the de-duplicated patient-level data; PACA-AU array duplicates excluded). Puleo is the largest contributor, supplying 288 of the 570 unique patients (51%), which is why the combined estimate is also reported with each dataset omitted in turn; those leave-one-dataset-out estimates range from 1.34 to 1.45. (b) Correspondence (|Spearman r|) of tuned supervised risk scores (penalized Cox and supervised PCA) with the three DeSurv programs (D1–D3); supervised scores concentrate on D1 and are only weakly correlated with the D2 stromal program. No validation-cohort survival outcomes were used in training, scoring or hyperparameter selection.

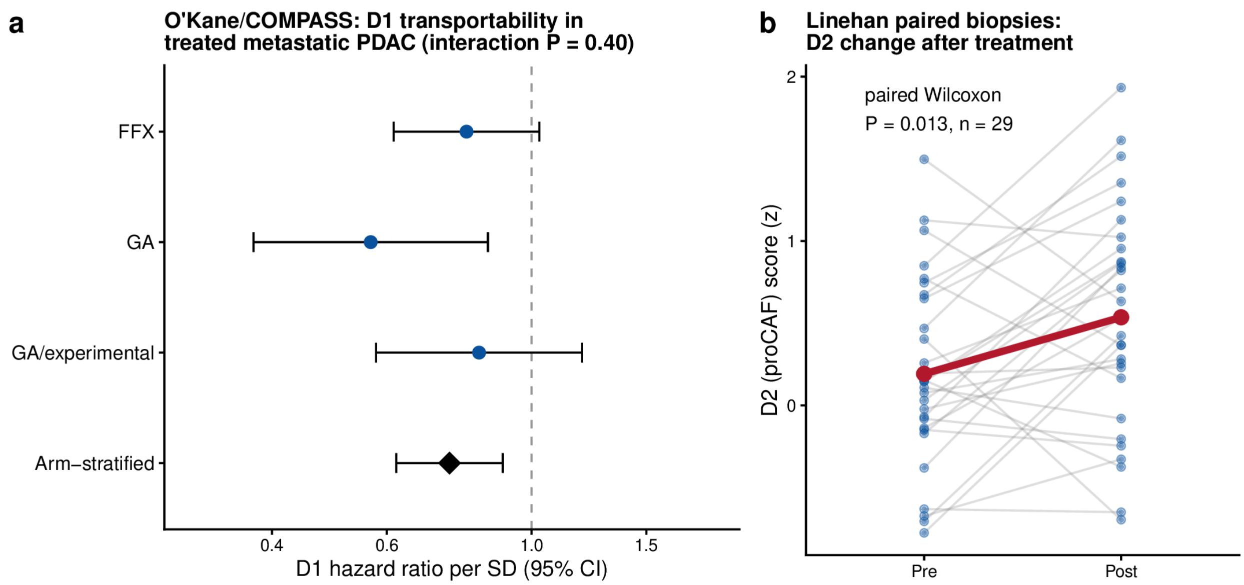

Figure 4: Transportability of the frozen DeSurv programs into independently treated PDAC cohorts. (a) Prognostic transportability of D1 in treated metastatic PDAC. Forest plot of hazard ratios per SD of the frozen D1 score within each O'Kane/COMPASS treatment arm (FOLFIRINOX, gemcitabine/nab-paclitaxel [GA], and GA/experimental); the diamond is the common association from an arm-stratified Cox model with treatment-specific baseline hazards (not a pooled-arm estimate), HR 0.75 (95% CI 0.62–0.90, $P = 0.003$). D1 was protective in all three arms; the arm-by-score interaction was not significant (interaction $P = 0.40$), though the individual arms are underpowered for detecting heterogeneity. (b) The proCAF program D2 before and after treatment in paired Linehan biopsies ($n = 29$, paired Wilcoxon $P = 0.013$); grey lines are individual patients, the red line the group mean (+0.34 SD). Error bars denote 95% confidence intervals. Because treatment, stage and cohort were not independently varied, these analyses establish prognostic and biological transportability rather than treatment-predictive effects.

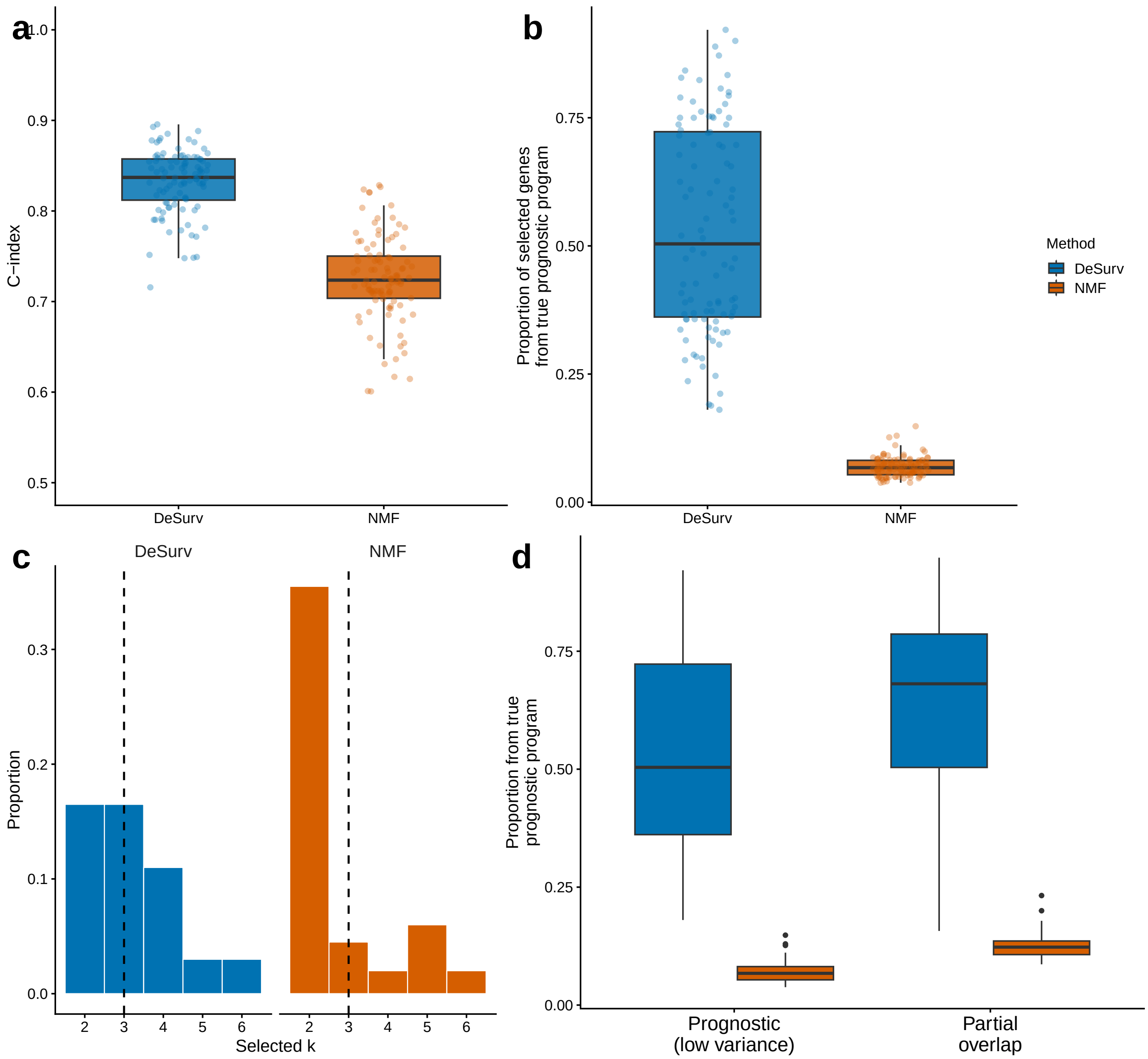

Figure 5: Survival supervision improves recovery of prognostic gene programs when variance and prognosis diverge. (a) Test-set concordance across 100 simulation replicates comparing DeSurv with unsupervised NMF ($\alpha = 0$; $p = 3{,}000$ genes, $n = 200$ samples, true $k = 3$; paired Wilcoxon $P < 0.001$). (b) Recovery of genes defining the true prognostic program, quantified as the proportion of selected genes in the best-matching learned program that belonged to the ground-truth prognostic gene set. In (a) and (b), boxes show median and interquartile range; whiskers extend to 1.5×IQR; points are individual replicates. (c) Distribution of selected ranks across replicates for DeSurv versus unsupervised NMF. (d) The same matched-factor precision metric as in (b), computed for two signal regimes: a prognostic program that explains low transcriptomic variance (separated from the dominant variance) versus one that partially overlaps the dominant variance. DeSurv recovers the prognostic program in both regimes, whereas unsupervised NMF, which follows variance, does not; the null scenario (no prognostic program) is shown in the Supplementary Information. These scenarios are constructed to isolate the regime in which the prognostic program carries low transcriptomic variance, and therefore quantify recovery under that construction; the concordance gap in (a) is not an expected discrimination gain in real data, where the two approaches perform comparably (Table 1).